```
#Write a program that takes a sentence as input and prints the number of vowels
sentence = input("Enter a sentence: ")

vowels = "aeiouAEIOU"
count = 0

for char in sentence:
    if char in vowels:
        count += 1

print("Number of vowels:", count)
print("Reversed sentence:", sentence[::-2])
print("Uppercase sentence:", sentence.upper())
```

```
Enter a sentence: Rohit
Number of vowels: 2
Reversed sentence: thR
Uppercase sentence: ROHIT
```

```
#Question 2 — Lists
# Write a program that takes a list of numbers and creates a new list containir
numbers = [1, 2, 3, 4, 5, 6, 7, 8]

squares = [x ** 2 for x in numbers if x % 2 == 0]

print("list:", numbers)
print("even numbers list:", squares)
```

```
list: [1, 2, 3, 4, 5, 6, 7, 8]
even numbers list: [4, 16, 36, 64]
```

```
# Question 3 — Tuples
#Write a program that stores the name and marks of 5 students as a list of tupl
students = [
    ("Rohit", 85),
    ("Amit", 92),
    ("jitendra", 78),
    ("Hari", 95),
    ("Rahul", 88)
]

highest_student = max(students, key=lambda x: x[1])

print("Student with highest marks:", highest_student[0])
print("Highest marks:", highest_student[1])
```

```
Student with highest marks: Hari
Highest marks: 95
```

```
# . Question 4 — Sets
# Write a program that takes two lists of student names (Section A and Section


section_a = ["Rohit", "Amit", "Hari", "jitendra"]
section_b = ["Amit", "Ram", "Rahul", "Kiran"]

A = set(section_a)
B = set(section_b)

print("Students in both sections:", A & B)

print("Students only in Section A:", A - B)

print("Students only in Section B:", B - A)
```

```
Students in both sections: {'Amit'}
Students only in Section A: {'jitendra', 'Rohit', 'Hari'}
Students only in Section B: {'Kiran', 'Ram', 'Rahul'}
```

```
#Question 5 — Dictionaries
# Write a program that stores at least 5 products wi
# Dictionary containing products and prices
products = {
    "book": 500,
    "Lunch box": 400,
    "Bag": 800,
    "Pen box": 50,
    "Laptop": 68000
}

print("Products with price greater than 100:")

for product, price in products.items():
    if price > 100:
        print(product)
```

```
Products with price greater than 100:
book
Lunch box
Bag
Laptop
```